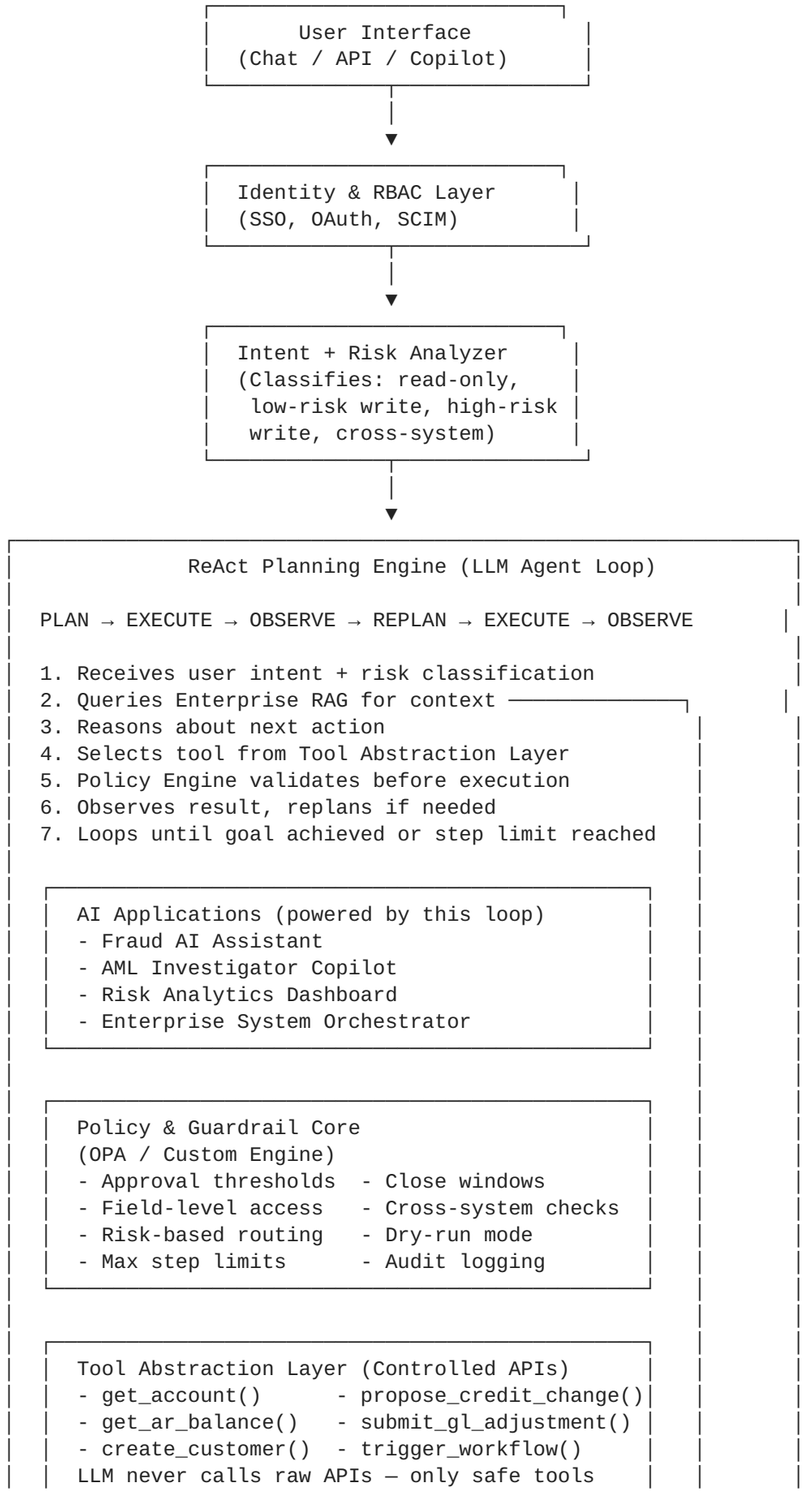
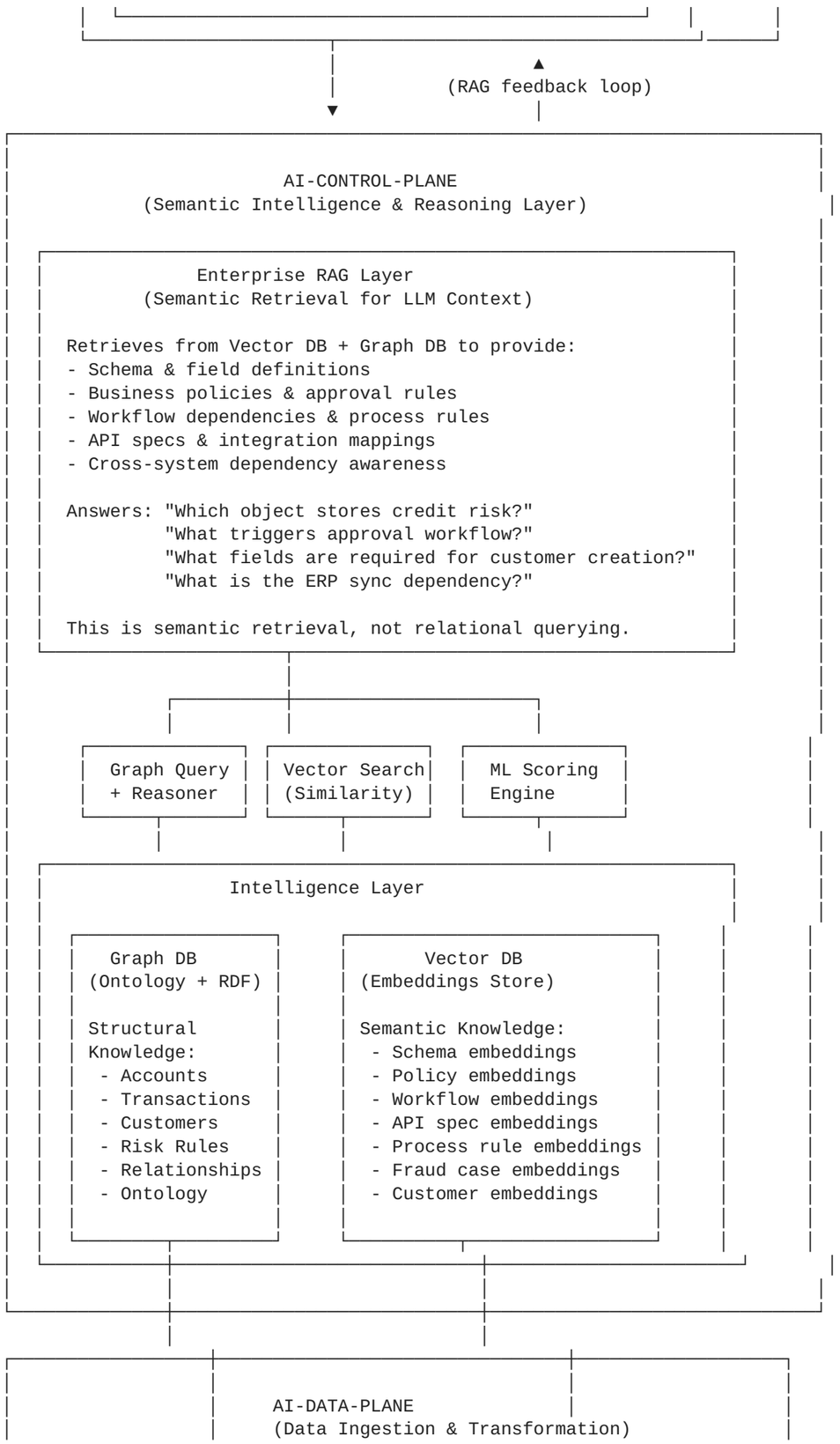
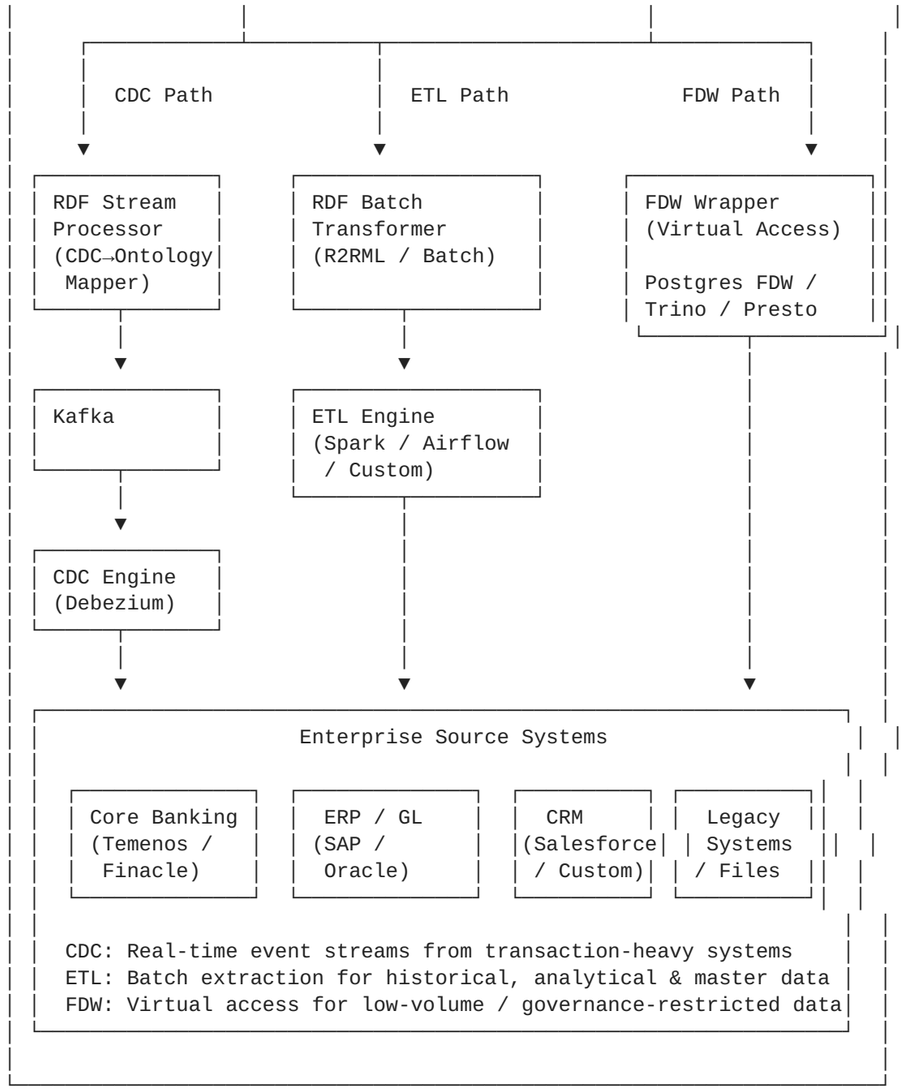
Total System Architecture




(RAG feedback loop)
AI-CONTROL-PLANE
(Semantic Intelligence & Reasoning Layer)
Enterprise RAG Layer
(Semantic Retrieval for LLM Context)
Retrieves from Vector DB + Graph DB to provide:
- Schema & field definitions
- Business policies & approval rules
- Workflow dependencies & process rules
- API specs & integration mappings
- Cross-system dependency awareness
Answers: "Which object stores credit risk?"
"What triggers approval workflow?"
"What fields are required for customer creation?"
"What is the ERP sync dependency?"
This is semantic retrieval, not relational querying.
Graph Query + Reasoner
Vector Search (Similarity)
ML Scoring Engine
Intelligence Layer
Graph DB (Ontology + RDF)
Structural Knowledge:
- Accounts
- Transactions
- Customers
- Risk Rules
- Relationships
- Ontology
Vector DB (Embeddings Store)
Semantic Knowledge:
- Schema embeddings
- Policy embeddings
- Workflow embeddings
- API spec embeddings
- Process rule embeddings
- Fraud case embeddings
- Customer embeddings
AI-DATA-PLANE
(Data Ingestion & Transformation)


CDC Path
ETL Path
FDW Path
RDF Stream Processor (CDC→Ontology Mapper)
RDF Batch Transformer (R2RML / Batch)
FDW Wrapper (Virtual Access)
Postgres FDW / Trino / Presto
Kafka
ETL Engine (Spark / Airflow / Custom)
CDC Engine (Debezium)
Enterprise Source Systems
Core Banking (Temenos / Finacle)
ERP / GL (SAP / Oracle)
CRM (Salesforce / Custom)
Legacy Systems / Files
CDC: Real-time event streams from transaction-heavy systems
ETL: Batch extraction for historical, analytical & master data
FDW: Virtual access for low-volume / governance-restricted data